Joshy Sir

02-1-25

→ Conduction - Transfer of heat from one point to another without transfer of mass.
Eg - heating a metal rod.

→ Convection - Transfer of heat from 1 point to another ~~with~~ when there is a temperature difference with transfer of mass.
Eg - Boiling of water.
But when there is only 1°C change in temp while boiling water, it will be considered as conduction not convection.
ie, small change in temp is considered as conduction.

→ Buoyancy force will be greater when temp is increased. That's why hot water moves up & cold water moves below during boiling.

→ Radiation - Transfer of heat without any medium.
Eg - Solar radiation.

→ Reyleigh's number - Ratio between buoyant force & viscous force.

$$\text{Reyleigh's number} = \frac{\text{buoyant force}}{\text{viscous force}}$$

→ $$\text{Reynold's number} = \frac{\text{inertia force}}{\text{viscous force}} = \frac{\rho V D}{\mu}$$

when temp ~~is to~~ difference is low, Reyleigh's number will be small.

→ Critical Reyleigh's number = at that value, ~~conduction~~ convection starts.

→ Specific heat - heat required to raise the temp. of unit mass by 1°C. → J/kgK.

→ Thermal diffusivity = $\alpha = \frac{k}{\rho c}$.

It is the rate at which heat disperse through an object.

→ Newtons law of cooling → rate of heat loss of a body $\propto$ temp difference b/w body & environment.

$\boxed{Q \propto \Delta T}$ → $Q \propto (T_s - T_\infty)$

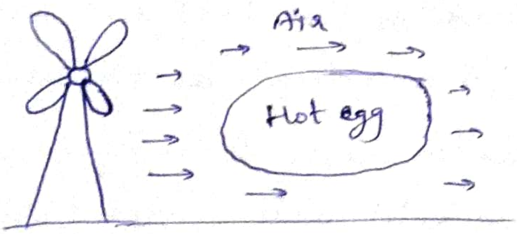
→ Conduction, $Q = -kA \cdot \frac{dT}{dx}$.

→ Convection, $Q = hA\Delta T = hA_s (T_s - T_\infty)$

where, h ⇒ convection heat transfer coefficient ⇒ W/m^2K

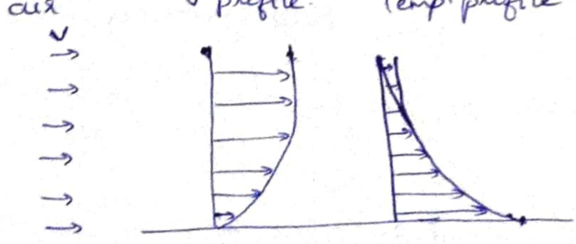
For forced convection, heat transfer, Q will be more ∴ h will be more.

→ Forced convection ⇒ cooling by a fan.



→ Natural convection ⇒ cooling by blowing of wind (natural process).
It is buoyancy driven.

— Consider a hot surface,



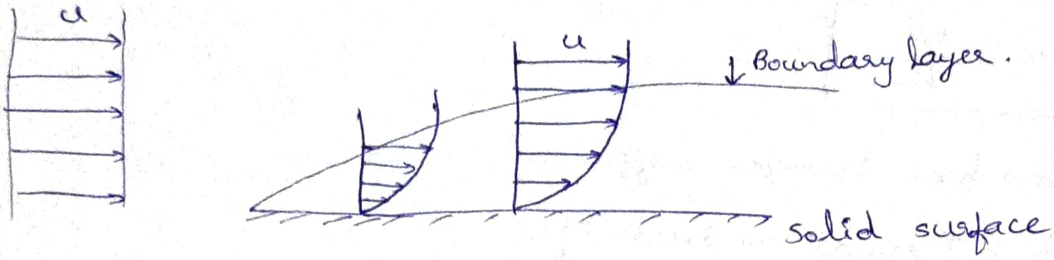
V ~~at~~ near surface will be zero due to no slip condition.

29-1-25. Heat capacity ⇒ Ability to contain heat. ~~J/kgk~~ ~~or~~

ρC_p ⇒ J/m^3k.

streams.

Consider a free ~~fluid~~ i.e, all layers flow with same velocity



There is no shear outside boundary layer. Inside the B.L. fluid layers travel with different velocity.

1. A 2m long, 0.3cm dia electrical wire extends across a room at 15°c. Heat is generated in the wire as result of resistance heating and the surface temperature of the wire is measured to be 152°c in steady operation. Also the voltage drop and electric current through the wire are measured to be 60V and 1.5 A, respectively. Disregarding any heat transfer by radiation, determine convection heat transfer coefficient for heat transfer b/w outer surface of the wire and the air in the room.

Ans. length of wire = 2m, $V = 60V$, $I = 1.5A$.

dia = 0.3 cm = 0.003m.

$$A = 2\pi r l = 2\pi \times \frac{0.003}{2} \times 2 = 0.0188 m^2.$$

$$Q = V \times I = 60 \times 1.5 = 90 \text{ W}.$$

∴ heat transfer coefficient, $h = hA(\Delta T)$.

$$\Delta T = 152 - 15 = 137°C = 410K.$$

∴ ~~$h = 90 \times 0.018 \times 410 = 664.2 \ w/m^2 k$~~.

$$h = \frac{Q}{A \Delta T} = \frac{90}{0.0188 \times 137} = 34.9 \ w/m^2 °k.$$

Nusselt number (Nu).

→ Dimensionless.

$$\boxed{Nu = \frac{hL}{K}}$$

h → heat transfer coefficient.

k → Thermal conductivity.

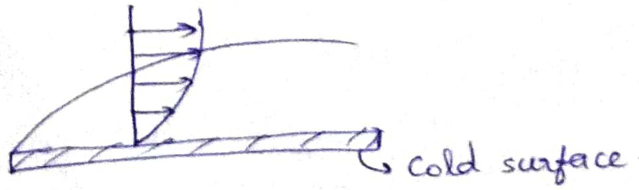
L → length.

$$q_{cond} = \frac{K\Delta T}{L}$$

$$q_{conv} = h\Delta T$$

$$\therefore Nu = \frac{q_{conv}}{q_{cond}} = \frac{h\Delta T L}{k\Delta T} = \frac{hL}{K}$$

Thermal boundary layer.



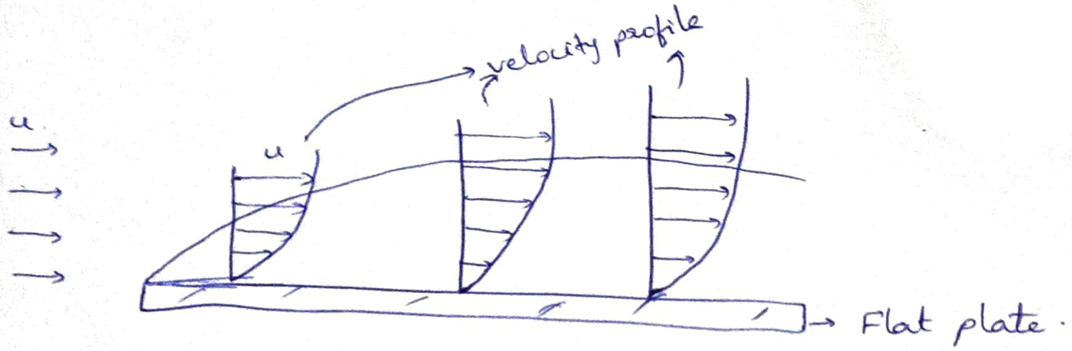
Prandtl number, $Pr = \frac{\delta v}{\delta t}$

ratio of thickness of velocity boundary layer & thermal boundary layer.

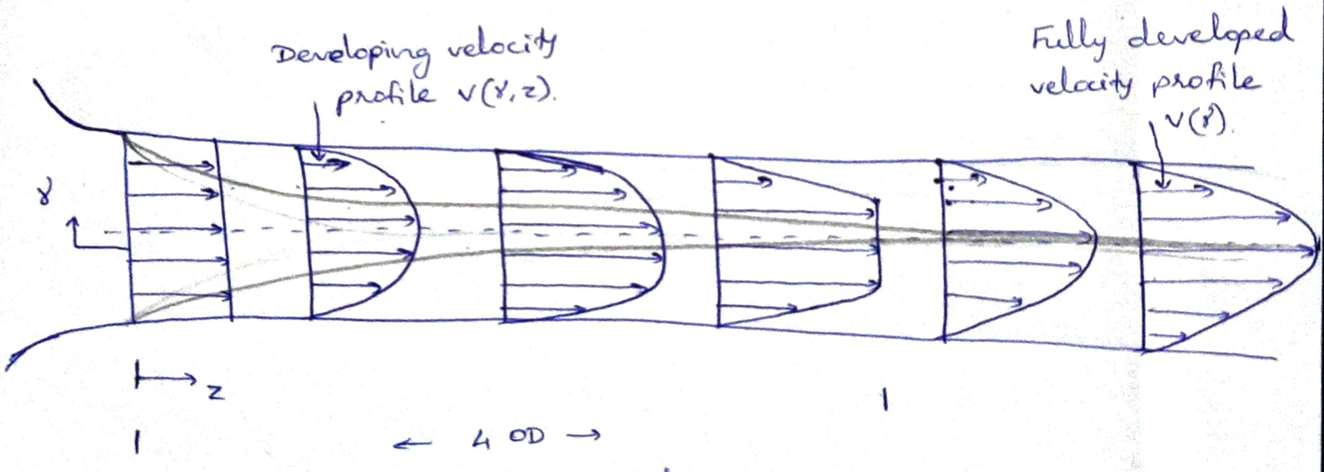
when thickness of velocity & thermal boundary layer is same, $Pr = 1$.

Rayleigh number = Grashoff number × Prandtl number.

$$\boxed{Ra = Gr \times Pr}$$



velocity boundary layer on a flat plate.



12/2/25

· Air at 20°c at at a pressure 1 bar is flowing over a flat plate at a velocity 3m/s. If the plate is 280mm wide and 56°C. Calculate following quantities at $x = 280mm$, given properties of air at bulk mean temperature ~~20+56~~ $\frac{20+56}{2} = 38°C$ are $\rho = 1.1374\ kg/m^3$, $K = 0.02732 W/m°C$, $C_p = 1.005 KJ/kgK$, $\nu = 16.768 \times 10^{-6} m^2/s$ $P_r = 0.7$

i) Boundary layer thickness ii) Local friction coe. iii) Avg friction coe. iv) Shearing stress due to friction. v) Thickness of thermal boundary layer. vi) Local convective heat transfer coefficient. vii) Average convective heat transfer coefficient. viii) Rate of heat transfer by convection. ix) Total drag force on the plate. x) Total mass flow rate through the boundary.

$$\delta = \frac{5x}{\sqrt{Re_x}} \qquad Re = \frac{3 \times 0.28}{16.768 \times 10^{-6}} = 5 \times 10^4$$

Since $Re_x < 5 \times 10^5$; hence flow is laminar. [For flat plate]

i) $\delta = \frac{5 \times 0.28}{\sqrt{5 \times 10^4}} = 6.26mm$

ii) local friction coefficient,

$$C_{fx} = \frac{0.664}{\sqrt{5\times10^4}} = \cancel{2.97\times10^{-3}}\ 0.003$$

iii) $$\bar{C_f} = \frac{1.328}{\sqrt{Re_x}} = 0.006$$

iv) $$\tau_0 = C_f \times \frac{\rho u^2}{2} = \frac{0.003\times1.1374\times3^2}{2} = 0.015\ N/m^2$$

v) Thickness of thermal boundary layer,

$$\delta_{th} = \frac{\delta}{(P_r)^{1/3}} = \frac{6.26}{(0.7)^{1/3}} = 7.05\ mm$$

vi) $$h_x = 0.332 \times \frac{k}{x} \times (Re_x)^{1/2} (P_r)^{1/3}$$

$$h_x = 0.332 \times \frac{0.02732}{0.28} \times (5\times10^4)^{1/2} \times (0.7)^{1/3} = 6.43\ W/m^2 C$$

vii) $$\bar{h} = 0.664 \times \frac{k}{L} \times (Re_L)^{1/2} (P_r)^{1/3}$$

$$= 0.664 \times \left(\frac{0.02732}{0.28}\right)(5\times10^4)^{1/2} \times (0.7)^{1/3} = 12.86\ W/m^2 C$$

viii) ~~Rate of~~ heat ~~transfer~~ Q

$$Q_{conv} = \bar{h} A_s \times (t_s - t_\infty) = 12.85 \times (0.28\times0.28)(56-20)$$
$$= 36.27\ W$$

ix) ~~Total~~ $F_D = \tau_0 \times Area = 0.01519 \times 0.28 \times 0.28 = 0.00119\ N$

x) Total mass flow rate through the boundary,

$$m = \frac{5}{8} \rho u (\delta_2 - \delta_1)$$ where $\delta_1 = 0$ at $x = 0$, $\delta_2 = \delta$ at $x = 0.28\ m$.

$$= \frac{5}{8}\, 1.1374 \times 3 \times (6.26\times10^{-3} - 0) = 0.0135\ kg/s$$

• Air at atmospheric pressure and 200°c flows over a plate with a velocity of 5m/s. The plate is 15mm wide and is maintained at a temp. of 120°c. calculate i) Thickness of hydrodynamic and thermal boundary layer. ii) local heat transfer coefficient at a distance of 0.5m from the leading edge. Assume that flow is on one side of the plate. $\rho = 0.815\ kg/m^3$, $\mu = 24.5\times10^{-6}\ Ns/m^2$, $Pr = 0.7$, $k = 0.0364\ W/m$.

$$\delta = \frac{5x}{\sqrt{Re_x}}$$

$$= 8.669\,mm$$

$x = 0.5\,m$

$$Re = \frac{\rho V D}{\nu} = \frac{\rho V D}{\mu} = \frac{0.415 \times 5 \times 0.5}{24.5 \times 10^{-6}}$$

$$= 83163$$

$\delta_{th} = 9.763\,mm$ $\quad h_x = 6.189\ W/m^2K$